AI Powered Document Analyzer Hub

Individual Project report

For the department of

Computer Science and Engineering (Artificial Intelligence and Machine Learning)/ Computer Science and Engineering (Artificial Intelligence)

Submitted By

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CERTIFICATE

This is to certify that the project titled Emotion recognition in conversations using Natural Language Processing submitted by **Rishav Poddar(University Registration No. 304202400900823 of 2024 - 2025),** student of the University of Engineering and Management, Kolkata, in partial fulfillment of requirement for the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning)/ Computer Science and Engineering (Artificial Intelligence), is a bonafide work carried out by them under the supervision and guidance of Prof. Debdatta Chatterjee during 4^{th} Semester of academic session of 2025 - 2026. The content of this report has not been submitted to any other university or institute. I am glad to inform that the work is entirely original and its performance is found to be quite satisfactory.

Signature of Supervisor
Department of CSE (AI & ML)/ Department of CSE (AI)

ACKNOWLEDGEMENT

We would like to express our profound gratitude to Prof. Debdatta Chatterjee, Project Supervisor, Prof. (Dr.) Sudipta Sahana, Head of the Department of CSE (AI & ML), and Prof. (Dr.) Sajal Dasgupta, Vice Chancellor, University of Engineering and Management, Kolkata for their contributions to the completion of my project titled Emotion recognition in conversations using Natural Language Processing .

We would like to express special thanks to our project mentor for his/her time and the efforts he/she provided throughout the semester. Your useful advice and suggestions were really helpful to us during the project's completion. In this aspect, we are eternally grateful to you.

We want to acknowledge that this project was completed entirely by ourselves and not by someone else.

Last but not least, we would like to extend our warm regards to our family members and peers who have kept supporting us and always had faith in our work.

Rishav Poddar and Signature with Date>>

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ABSTRACT

The digital information is growing fast and that is why we need systems that can look at the information and find the important parts. The old way of searching with keywords does not always work because it does not understand what the words mean in a sentence. So we made a system that uses intelligence to look at documents and find the right information.

This system breaks down the documents into pieces and looks for the important things in each piece. Then it makes a kind of map that shows how all the pieces are related to each other. We store these maps in a database so we can find the information again later. When someone asks a question the system turns the question into a map. Compares it to the maps we already have. This way it can find the relevant information.

The system also uses language models to make sure the answers are accurate and make sense. We designed the system to handle a lot of work like uploading documents preparing them for analysis making the maps storing them and answering questions. The system is strong and reliable. Can handle a lot of information which makes it good for things like research managing documents for companies and searching for information.

We found out that our system is much better at finding the information than the old ways. It uses the artificial intelligence techniques and has a strong foundation, which makes it a good solution for analyzing documents. Maybe one day we can make it even better, by adding pictures and other kinds of information and making it faster.

1.0 INTRODUCTION

1.1 Overview of Document Analysis Systems

We are living in a time where a lot of text is being created every day in areas like education, business, healthcare and research. This text is stored in documents like reports, research papers, emails and contracts.. It is getting very hard for people and organizations to manage and understand the information in these documents.

Old document analysis systems mostly relied on people reading the documents or using search tools that looked for specific words. But this method has a problem. It does not understand what the text really means. So people may not find the information they need if it is written in a way. This becomes an issue when there are many documents to go through because it takes a lot of time and can be wrong.

To solve these problems new computer techniques have been developed. These techniques are part of something called Natural Language Processing, which allows machines to understand and work with language. Document Analysis Systems that use Natural Language Processing can look at a lot of text. Find useful information on their own. With the development of intelligence Document Analysis Systems have changed from just looking for keywords to actually understanding what the text means.

Modern Document Analysis Systems use methods like taking text out breaking it down into parts making it into numbers and searching for meaning. These systems turn text into numbers so machines can see how different texts are related. This helps find information accurately and supports things like making summaries answering questions and searching in context. Document Analysis Systems can do all these things because they can understand what the text is really saying. Document Analysis Systems are getting better, at finding the information people need from the documents.

1.2 Role of AI in Document Understanding

Artificial Intelligence (AI) plays a crucial role in transforming traditional document processing into an intelligent and automated system. By leveraging machine learning and deep learning techniques, AI enables systems to analyze large volumes of unstructured text data efficiently and accurately. One of the key advancements in this field is the development of transformer-based models such as BERT, which are capable of understanding the context and semantics of language more effectively than earlier approaches.

In AI-powered document analysis, text is first processed and converted into embeddings, which are numerical vector representations capturing the semantic meaning of words, sentences, or entire documents. These embeddings allow the system to perform similarity-based comparisons, enabling semantic search instead of traditional keyword matching. This means that even if the user query does not exactly match the words in the document, the system can still retrieve relevant information based on meaning.

The integration of frameworks such as LangChain and pre-trained models from Hugging Face Transformers has further simplified the development of intelligent document analysis systems. These tools allow developers to build pipelines that include text extraction, embedding generation, and retrieval-based question answering. In the proposed system, embeddings are stored in a PostgreSQL database, enabling efficient retrieval of relevant information through similarity search.

The AI Powered Document Analyzer Hub leverages these technologies to provide an efficient solution for document analysis. It allows users to upload documents, process their content, and interact with them through intelligent queries. The system reduces manual effort, improves accuracy, and enhances the overall efficiency of information retrieval. By incorporating AI techniques, the proposed system demonstrates a significant advancement over traditional document processing methods and provides a scalable solution for real-world applications.

1.3 Applications of Artificial Intelligence in Document Analysis

Artificial Intelligence is really changing how we look at documents and use the information in them. Artificial Intelligence can look at a lot of text understand what it says and make sense of it. This is very helpful in areas.

In schools and when people are doing research Artificial Intelligence helps students and researchers read papers and find the important parts. This saves a lot of time. Helps people get more work done. Companies also use Artificial Intelligence to look at reports, contracts and emails. This helps them make decisions and work more efficiently.

Artificial Intelligence is also used in law and money matters to look at documents and make sure everything is okay. It can even find patterns that do not seem right which helps catch cheating and reduce risks. In hospitals Artificial Intelligence helps doctors and nurses look at records and medical reports. This helps them find information quickly.

New systems that look at documents use something called Natural Language Processing to understand what the words mean. They use tools like LangChain and Hugging Face Transformers to do things like summarize documents put them into groups and answer questions. This makes them very good at what they do.

The Artificial Intelligence Powered Document Analyzer Hub is a system that brings all these things together. It lets people upload documents and work with them in a way. This makes it very useful in areas and shows how important Artificial Intelligence is in how we work with documents now. Artificial Intelligence in document analysis is very helpful. Makes things easier, for people who work with documents.

2. LITERATURE SURVEY :

2.1 Overview of Document Analysis using NLP:

Document analysis has changed a lot with improvements in Natural Language Processing techniques. NLP helps machines understand and process language in a meaningful way. Earlier systems used text processing methods like breaking down text into words removing common words like "the" and "and" and analyzing word frequencies. These methods were okay for tasks but couldn't grasp the context and meaning of the text.People want systems so document analysis now focuses on understanding the meaning of text. New approaches try to get an understanding of text letting systems see how words, sentences and documents relate to each other.

These techniques are used in things like summarizing texts finding information analyzing opinions and answering questions. Recently AI has brought in models that can handle large amounts of unstructured data.These models can learn relationships, within text leading to accurate results. As a result NLP is now a part of creating intelligent document analysis systems. Document analysis relies heavily on NLP to understand and process text. NLP enables machines to make sense of language.

2.2 Traditional Keyword-Based Search Approaches

Traditional document retrieval systems primarily rely on keyword-based search methods. These systems match user queries with exact words or phrases present in the documents. Techniques such as Boolean retrieval and term frequency-based ranking were commonly used to identify relevant documents.

One widely used method is TF-IDF (Term Frequency–Inverse Document Frequency), which assigns importance to words based on their frequency in a document relative to their occurrence across multiple documents. Although keyword-based approaches are simple and computationally efficient, they have several limitations. They fail to capture the semantic meaning of the text and are unable to identify relationships between different words that convey similar meanings.

For example, a search query containing the word “automobile” may not retrieve documents containing the word “car,” even though both have similar meanings. This limitation reduces the effectiveness of traditional search systems, especially when dealing with large and diverse datasets.

2.3 Classical Machine Learning Methods

To overcome the limitations of keyword-based approaches, classical machine learning techniques were introduced in document analysis. Algorithms such as Naive Bayes, Support Vector Machines (SVM), and Logistic Regression have been widely used for tasks like text classification, spam detection, and sentiment analysis.

These methods typically rely on feature extraction techniques such as bag-of-words and TF-IDF representations. While machine learning models improve performance compared to traditional methods, they still depend heavily on manual feature engineering and lack deep contextual understanding of language.

Moreover, classical models struggle with handling large-scale data and complex linguistic patterns. They are not capable of capturing long-range dependencies within text, which limits their effectiveness in advanced document analysis tasks.

2.4 Deep Learning Approaches (RNN / CNN)

The introduction of deep learning marked a significant advancement in the field of document analysis. Neural network-based models such as Recurrent Neural Networks (RNN) and Convolutional Neural Networks (CNN) have been used to process sequential and structured text data more effectively.

RNN-based models are particularly useful for handling sequential data, as they maintain information about previous inputs while processing new data. This allows them to capture contextual relationships within text. However, RNNs suffer from issues such as vanishing gradients, which limit their ability to handle long sequences.

CNN-based models, on the other hand, are effective in extracting local features and identifying important patterns within text. They are computationally efficient and have been successfully applied to tasks such as sentence classification and document categorization.

Despite their advantages, deep learning models still have limitations in capturing global context and require large amounts of training data. This led to the development of more advanced models capable of understanding context more effectively.

2.5 Transformer-Based Models (BERT, RoBERTa, DistilBERT)

Transformer-based models have revolutionized the field of NLP by introducing mechanisms that allow models to focus on different parts of the input text simultaneously. One of the most influential models is BERT, which uses bidirectional encoding to understand context from both directions of a sentence.

Variants such as RoBERTa and DistilBERT further improve performance by optimizing training techniques and reducing model size while maintaining accuracy. These models generate contextual embeddings, which capture the meaning of words based on their surrounding context.

Transformer-based models have significantly improved the performance of tasks such as question answering, text summarization, and semantic search. They form the foundation of modern document analysis systems and enable more accurate and efficient processing of text data.

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2.7 Multi-task, Transfer Learning and Domain Adaptation:

Recent works leverage multi-task learning (predicting multiple emotion labels, sentiment, and intensity jointly), and domain adaptation techniques (fine-tuning on target domain chat logs or using adversarial training) to close the gap between general datasets (Reddit/Twitter) and domain-specific conversations (customer support, healthcare). Pretraining on large general corpora followed by careful domain-specific fine-tuning remains the best practical approach.

2.8 Retrieval Augmented Generation Systems

Retrieval Augmented Generation Systems are a way of doing things. It is a way that combines finding information with making text. First the system finds the information it needs from a database. It uses techniques to search for the information. Then a language model makes a response based on what it found.

The Retrieval Augmented Generation Systems make the responses more accurate. They do this by using information. This way of doing things is used in lots of places. For example it is used in chatbots and virtual assistants. It is also used in systems that answer questions, about documents.

The AI Powered Document Analyzer Hub works in a way. It. Finds document information using a special kind of search. This search is called search. The system can give responses to what the user asks. It does this by combining the search and the generation of text. The Retrieval Augmented Generation Systems make sure the responses are meaningful and make sense. The Retrieval Augmented Generation Systems are very useful.

3.1 Problem Definition

We live in a time where organizations and individuals create and store a lot of information in documents like PDFs, reports and emails. It is getting harder to manage and find information in these documents because there is just so much of it. The old ways of dealing with documents involve people reading them and searching for words, which takes a long time and is not very efficient.

When we search for something using keywords the system looks for matches in the documents. This is a way to do things but it does not understand what the words really mean. This means we often get search results that're not accurate. For example if we search for "data analysis techniques" we might not find documents that talk about "data processing methods" or "analytical approaches" even though they are related.

Another big problem with systems is that they cannot handle a large number of documents very well. As we get more documents it becomes harder to find what we need. It takes a time and is prone to mistakes. This is not good enough for things that need to be accurate and quick.

Also the systems we have now are not very smart. They cannot understand what we mean when we ask a question or find connections between pieces of text. This makes it hard for us to get information, from big datasets, which can affect how well we work and make decisions.

Now that Artificial Intelligence is getting better with Natural Language Processing we need systems that can understand documents in a smarter way. Document analysis systems should be able to understand what the text means handle a lot of data and give us results that make sense. We need these systems to deal with the amounts of Document analysis and data that we have. The Document analysis systems should be able to process the data and provide us with accurate Document analysis results.

3.2 Scope of the Problem

The issue of document analysis is a big problem that affects many areas where a lot of text data is created and used every day. These areas include research, business organizations, legal systems, healthcare and information technology. In each of these areas users need to find the information they need from big collections of documents quickly and correctly.

In research settings students and researchers have to read a lot of research papers, articles and reports to get the information they need. Reading all these documents by hand takes a time and they might miss important details. It is the same in business organizations, where a lot of data is stored in reports, emails and contracts that need to be analyzed to make decisions and plan ahead.

In the field professionals work with complex documents like case files, agreements and contracts. Finding parts or information in these documents by hand can be boring and prone to mistakes. In healthcare patient records and medical reports have information that needs to be found quickly and correctly. A good document analysis system can make a difference in how fast and well decisions are made in these situations.

The system we are talking about is designed to solve these problems by creating a document analysis platform that uses intelligence. The system is made to work with text-based documents find the information and provide smart search options. It uses methods like creating embeddings and semantic similarity to allow for context-aware information retrieval.

The document analysis system uses technologies like FastAPI for the backend and PostgreSQL for storing and managing embeddings. The system lets users upload documents turn them into organized representations and do searches based on similarity to find the information they need quickly.

However the current system only works with text-based documents. It does not fully support things like images, audio or video files. Also dealing with documents, in languages and complex data formats is still a challenge and could be improved in the future. The document analysis system needs to be able to handle these things to be really useful. The document analysis system is a tool that can help people in many areas.

3.3 Challenges in Existing Systems

There are problems with the systems we have now for handling documents. Even though we have a lot of tools for processing documents we still have some issues. One of the problems is that the search systems that use keywords do not really understand what people are looking for. These systems cannot figure out what the person wants so they give us results that're not what we need or they do not give us all the information.

Another big problem is that these systems can become slow when we have a lot of documents. When the number of documents gets really big the old systems start to have trouble. We need ways to store organize and find documents quickly which the old ways cannot do.We also have a lot of trouble when we try to get information from documents. Sometimes documents have mistakes extra information we do not need or they are not formatted well. This can make it hard to get the information from them. We need ways to clean up the documents and get the information we need. Making embeddings, which are like maps of what the documents are about and searching for similar documents quickly can take a lot of computer power. We also need to be able to store and get these embeddings quickly which is another challenge. We can use databases, like PostgreSQL to store them. We still need to make sure they work well with a lot of documents.

All these challenges show that we need a system that can handle documents in a smart way understand what people are looking for and work well even with a lot of documents. We need document processing systems that're advanced and can work with a lot of documents and document analysis systems that can understand what people are looking for and systems that can handle a lot of documents, which are the document systems.

4.1 Problem Solution

The AI Powered Document Analyzer Hub is a system that helps with ways of looking at documents. It uses ideas from Natural Language Processing to make things better. This system is different from others because it does not just look for keywords it tries to understand what the words really mean. This helps people find the information they need easily and accurately.

When you put a document into the system it changes the document into a form that's easy to look at. The system takes the words out of the document. Breaks them down into smaller parts. These parts are then turned into numbers that show what the words mean. The AI Powered Document Analyzer Hub uses these numbers to help people find information even if they use different words. This is really helpful because people can find what they need even if they do not know the words to use.

The AI Powered Document Analyzer Hub makes it so people do not have to look at documents by hand much. It helps people work with documents in a natural way. They can ask the AI Powered Document Analyzer Hub questions. Get exact answers based on what is in the documents. This makes it faster and more accurate to get information from a lot of data. The AI Powered Document Analyzer Hub is very useful, for the AI Powered Document Analyzer Hub.

4.2 Overview of the Proposed System

The system for solving the problem is like a series of steps that work together. This system starts with the user uploading documents to the interface. The documents that are uploaded are then looked at closely to get the text out of them. This text is then cleaned up to get rid of things that are not needed like formatting that does not look right and other noise. This step makes sure that the text is quality and ready for the next steps.

After the text is cleaned up it is broken down into pieces so it can be handled and looked at more easily. Each small piece is then turned into something called an embedding using models. These embeddings show what the text really means and are used for searching. The embeddings that are made are stored in a database using PostgreSQL, which's good for storing and getting back special kinds of data.

When a user asks a question the system looks at it in a way by turning it into an embedding. This embedding of the question is then compared to the embeddings of the documents that are stored. The system looks for the text that's most relevant and gets it based on how close it is to the question. The text that is gotten back is then used to make a response that's meaningful and takes into account the context. This means the user gets the information quickly.

The backend of the system which's the part that the user does not see is made using FastAPI. This makes the system work well and handle many requests at the same time. The system uses tools and databases which means it can handle a lot of data and is scalable. The document processing and analysis system is very good, at handling datasets and providing accurate information. The document analysis system is made to be efficient and provide results.

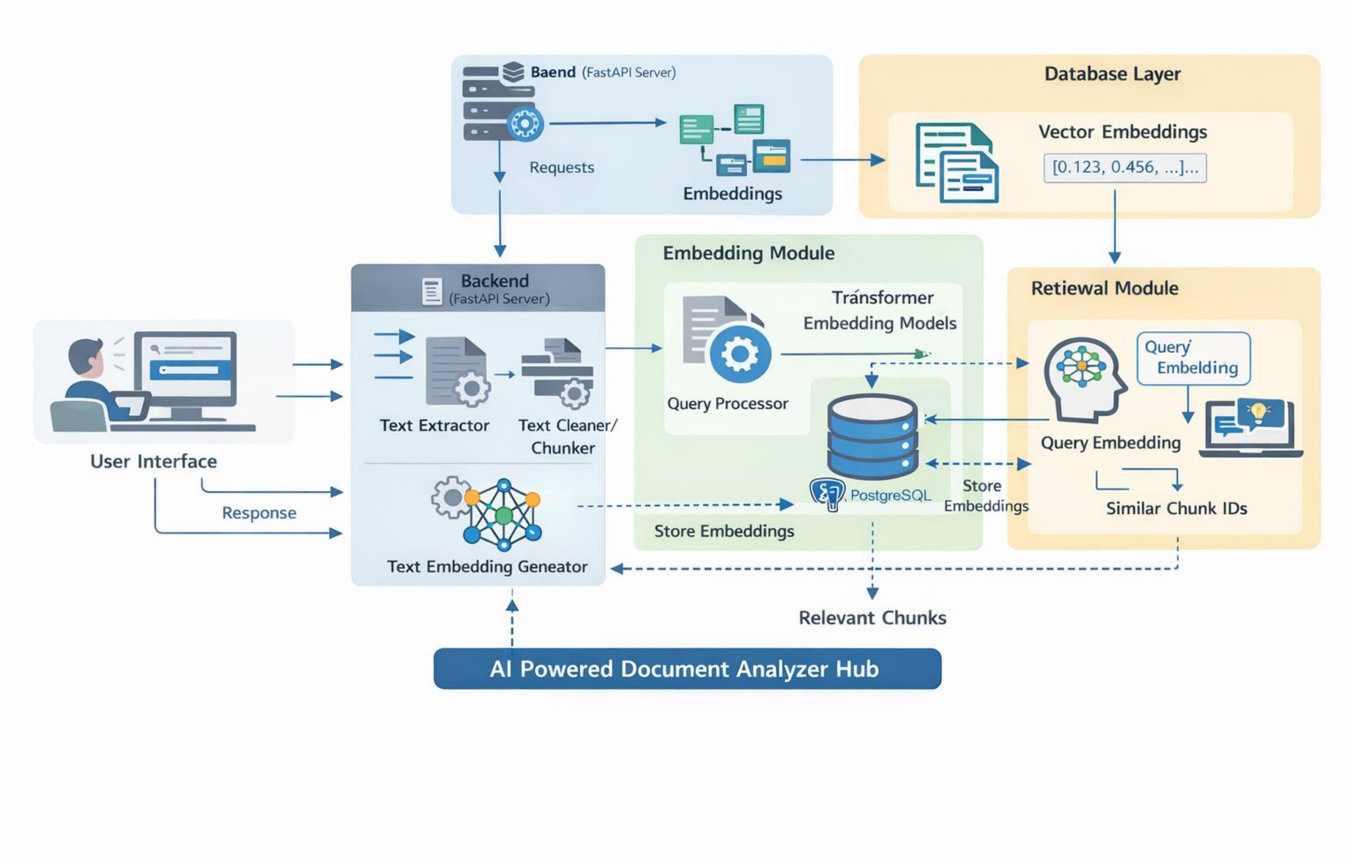

Image 1.1 (System Architecture)

4.3 System Architecture and Working

The architecture of the proposed system is designed to ensure smooth interaction between different components involved in document analysis. The system consists of multiple layers that work together to process user inputs and generate outputs efficiently.

At the front end, users interact with the system by uploading documents and submitting queries. These requests are handled by the backend API, which is responsible for coordinating all operations within the system. The backend, built using FastAPI, manages tasks such as document processing, embedding generation, and query handling.

The document processing module plays a crucial role in extracting and preparing text from uploaded files. It ensures that the data is cleaned and divided into manageable chunks. These chunks are then passed to the embedding module, where they are converted into vector representations. The embeddings are stored in PostgreSQL, which acts as the central storage system for both text data and their corresponding vector representations.

When a query is received, the retrieval module converts it into an embedding and performs a similarity search against the stored embeddings. This process identifies the most relevant pieces of information from the document. The response generation module then uses these results to construct a meaningful answer, which is returned to the user.

The architecture is designed to be modular and scalable, allowing additional features such as image processing or multilingual support to be integrated in the future without major modifications.

4.4 Data Flow Summary:

- User uploads document → API receives → stored to S3 + metadata saved → enqueue ingestion job.

- Worker extracts & cleans text → chunks → embeddings → vector DB.
- When user asks a query: embed query → retrieve chunks → LLM produces answer.
- If user uploads chat logs or uses the chat interface: emotion classifier processes text → outputs emotion data → UI shows analytics.
- Everything is multi-tenant (per organisation), secure, and isolated.

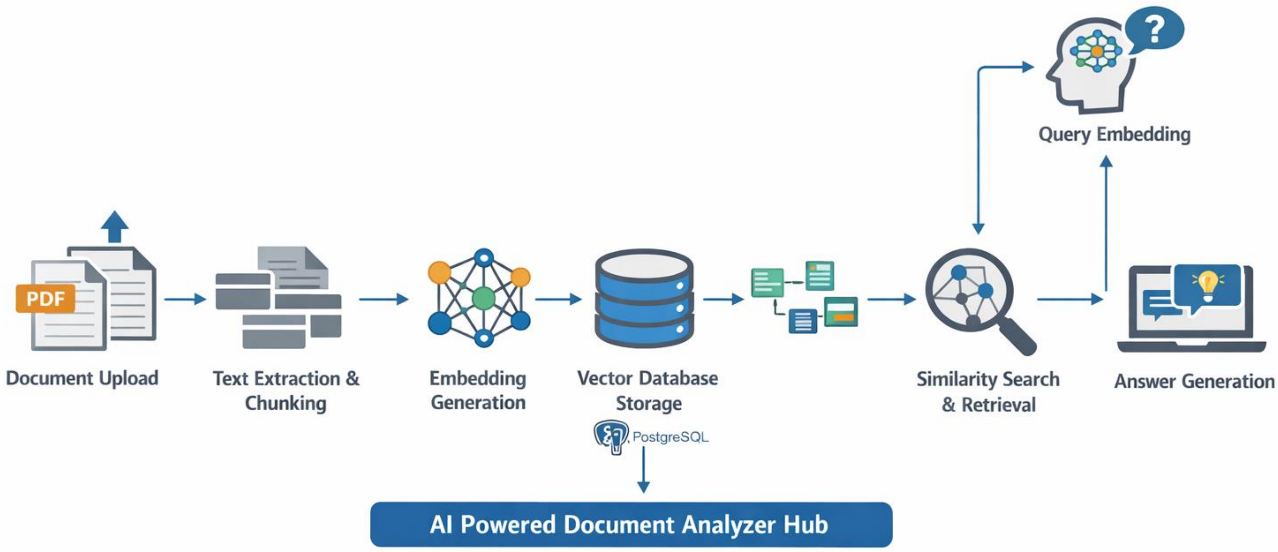

Image 1.2 (Data Flow Diagram)

4.5 Expected Outcome

The proposed system is expected to significantly improve the efficiency and effectiveness of document analysis. By leveraging semantic search and embedding-based retrieval, the system enables users to access relevant information quickly and accurately. It reduces the need for manual reading and simplifies the process of extracting insights from large volumes of data.

The system also enhances user experience by allowing natural language interaction with documents. Users can ask questions in a conversational manner and receive precise answers without needing to search through entire documents. This makes the system particularly useful in environments where quick access to information is critical.

Overall, the AI Powered Document Analyzer Hub provides a scalable and intelligent solution for document processing. It demonstrates how modern technologies such as **Natural Language Processing**, **FastAPI**, and **PostgreSQL** can be integrated to build an efficient and user-friendly system for analyzing unstructured data.

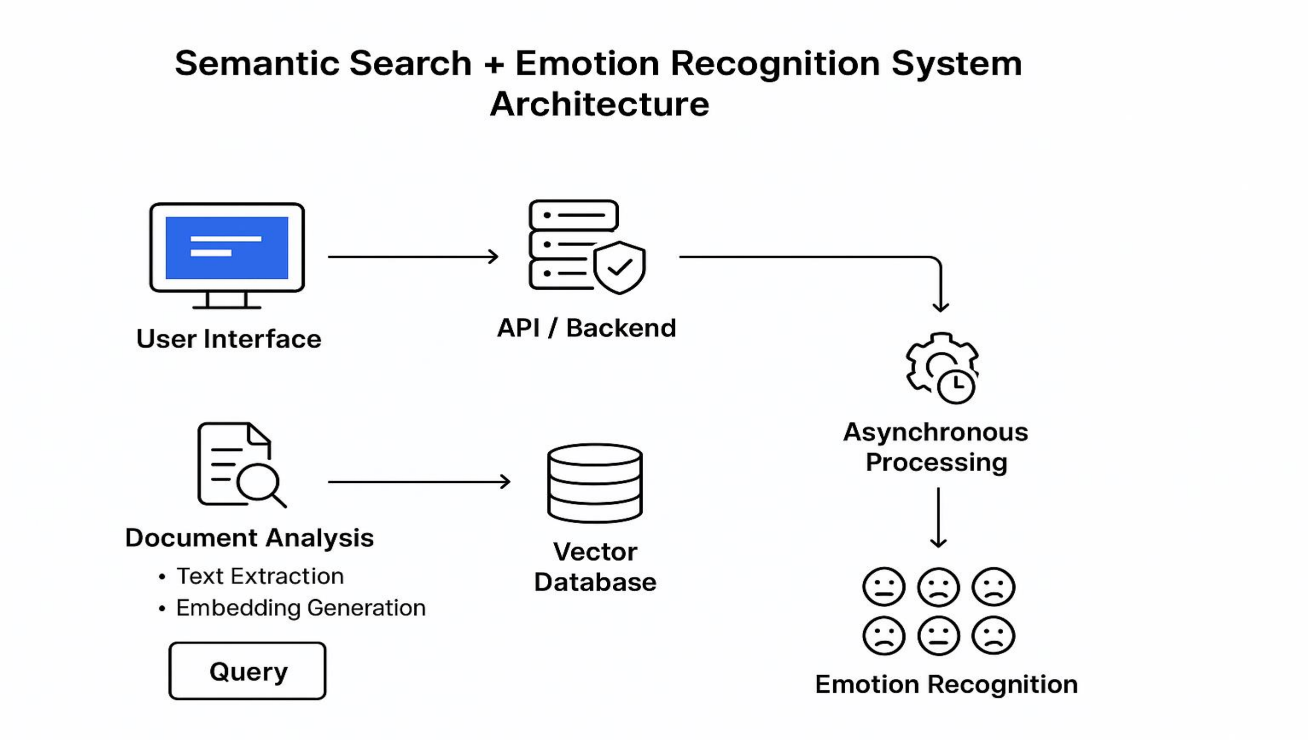

Image : 1.3(overall Architecture)

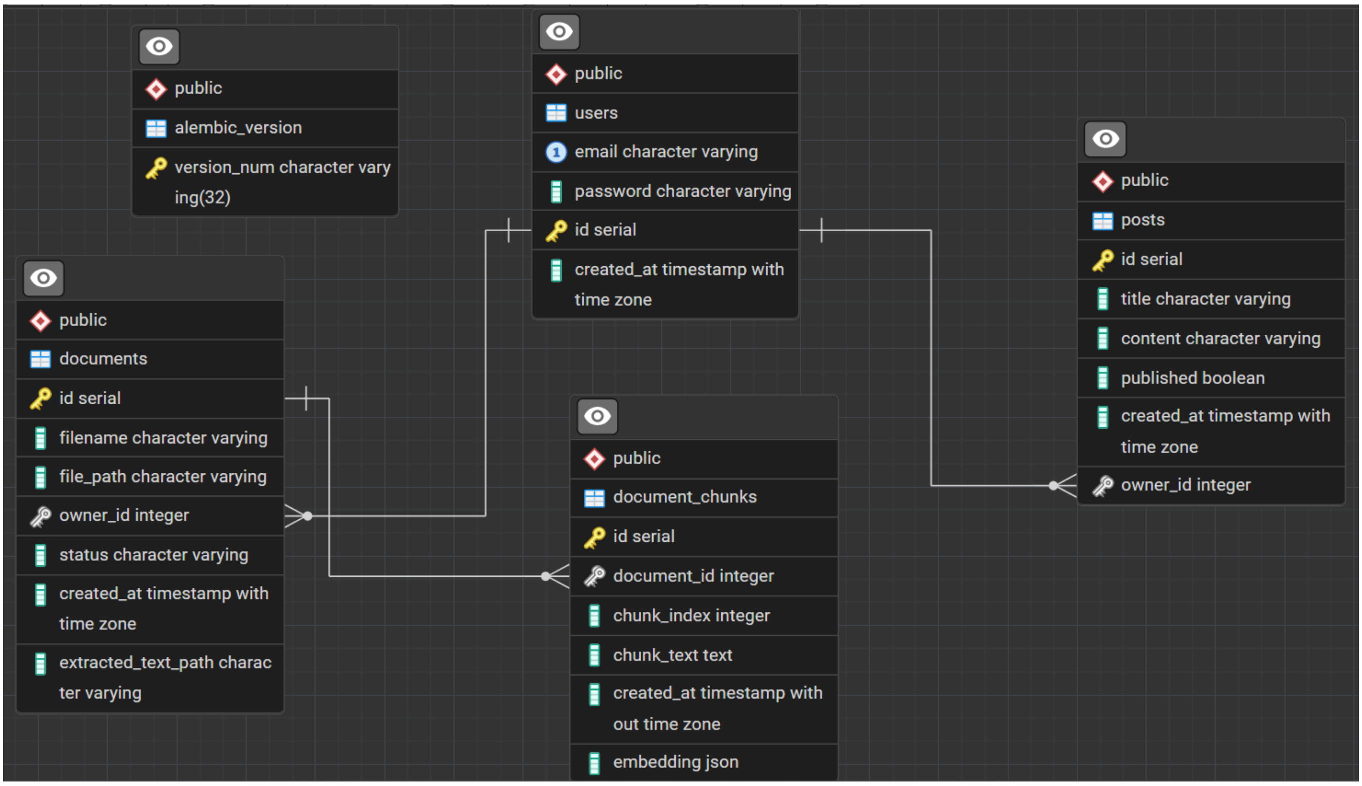

Image 1.4 (ER Diagram of a Database)

5: EXPERIMENTAL SETUP :

5.1 Experimental Setup

5.1 Experimental Setup

The AI Powered Document Analyzer Hub is tested to see how well it works with documents. The system uses technologies and artificial intelligence to process and retrieve information.

The system is built using FastAPI, which handles API requests quickly. This means it can deal with user requests at the same time without slowing down. The information from documents is stored in PostgreSQL, which manages data and details.

The system runs on a computer with a multi-core processor, enough RAM and local storage. This shows that it does not need a powerful computer and can work on typical machines.

Here is how the test works:

Documents are uploaded to the system.

The system extracts text, from the documents. Breaks it into smaller parts.

Each part is converted into a code using transformer-based models.

The codes are stored in the database. Used to find similar information when a query is made.

The system is judged on how it finds relevant information and how fast it does so. Test queries are given to the system and its responses are checked for accuracy, relevance and response time.

The AI Powered Document Analyzer Hub is evaluated based on its ability to retrieve information accurately and efficiently.

The AI Powered Document Analyzer Hub uses FastAPI and PostgreSQL.

The AI Powered Document Analyzer Hub is tested with test queries.

The system responses generated are analyzed for correctness, relevance and response time.

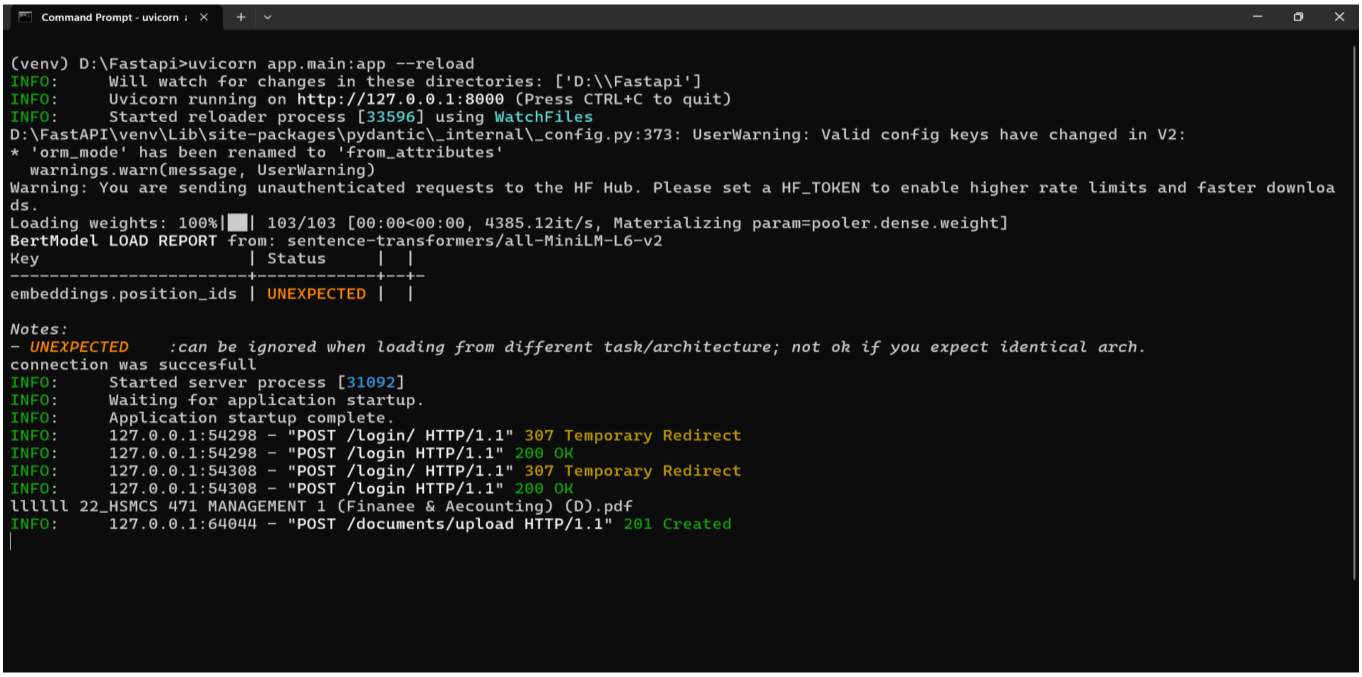

Image 1.5(Server Running in Localhost)

5.1.1 Dataset Description

The dataset we are using for this project is a collection of text-based documents. These documents include things like PDFs, academic papers and regular text files. The documents have a lot of text in them which we use as the input for our document analysis system.

Our dataset is not about one specific topic. It has all kinds of things in it so we can test our system with different types of content. This way we can see if our system can handle real-life situations where documentsre all different in terms of how they are set up how long they are and how complicated they are. We clean up the documents by taking out formatting and noise so we are left with just the plain text.

We break each document into pieces so our system can work with them more easily. Breaking them up like this really helps our system get the answers because it can focus on just a small part of the text at a time. We take these pieces and turn them into something called embeddings and then we store them in a database called PostgreSQL.

We use the dataset to test things about our system like how it takes in documents makes embeddings searches for similar things and generates answers. By using a dataset with different things in it we can see how well our system works and if it can adapt to new things. The dataset helps us make sure our system is good, at handling all kinds of documents and that it is strong and can work well in situations.

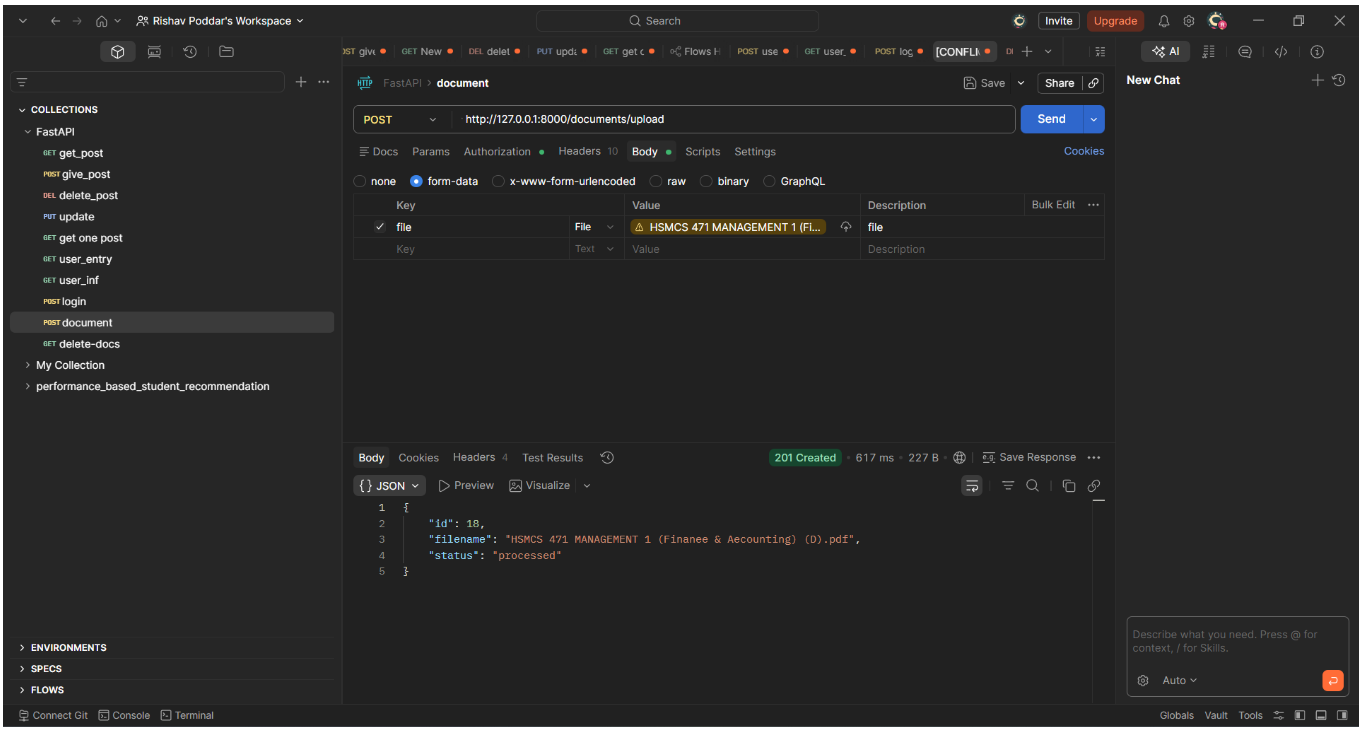

Image 1.6(Uploading Document)

5.2 Result Analysis

The AI Powered Document Analyzer Hub works well. We looked at how it does in a few areas like accuracy how long it takes to respond and how relevant the results are. This system is a lot better than the way of searching with keywords. It gives us results that make sense because it understands the context.

When we tested it the system found the relevant parts of documents when we asked it questions. It could even find the parts when our question was not exactly the same as the words in the document. This is because it uses something called embeddings to find meanings. This shows that Natural Language Processing is really good at helping us find the information we need.

The system also responds quickly. It uses something called FastAPI to handle requests. When it looks for things in the database it gives us results fast. This makes it good for things that need to happen in time.

The system is also good at finding the information in documents.. How well it does depends on things like how good the embeddings are and how we prepare the documents. If we adjust these things the system will work better.

The results show that the AI Powered Document Analyzer Hub is really good, at looking at documents and giving us information. It saves people a lot of time and effort when they have to review documents. This makes people more productive when they use the AI Powered Document Analyzer Hub.

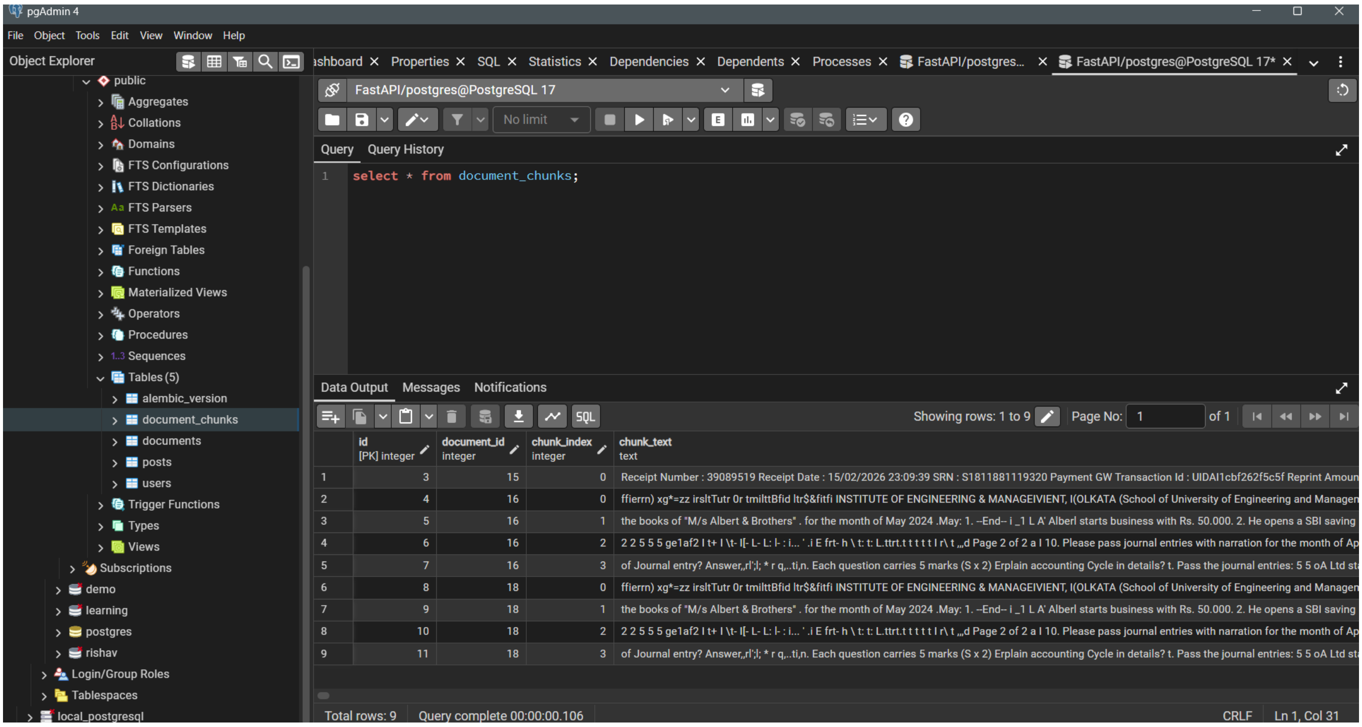

Image 1.7(Document chunk is getting saved in database)

5.3 Performance Evaluation

The system is evaluated using qualitative and quantitative measures. Qualitatively, the responses generated by the system are assessed based on their relevance and coherence. Quantitatively, performance can be measured using metrics such as response time and retrieval accuracy.

The system shows strong performance in semantic retrieval tasks, where it outperforms traditional search methods. The use of embeddings allows the system to understand the context of both documents and queries, leading to more accurate results. Additionally, the scalable architecture ensures that the system can handle increasing amounts of data without significant performance loss.

5.4 Limitations We Found During Testing

We noticed some problems with our system while testing. One big issue is dealing with documents that have pictures or scanned text. This is hard because our system is mainly designed for text data. Also how well our system works depends on the quality of the embeddings it creates. Different models can produce embeddings of qualities.

Another problem is that creating embeddings can be expensive in terms of computer power, for huge datasets. Our system works fine with an amount of work but it might need more optimization if we want to use it on a really large scale.

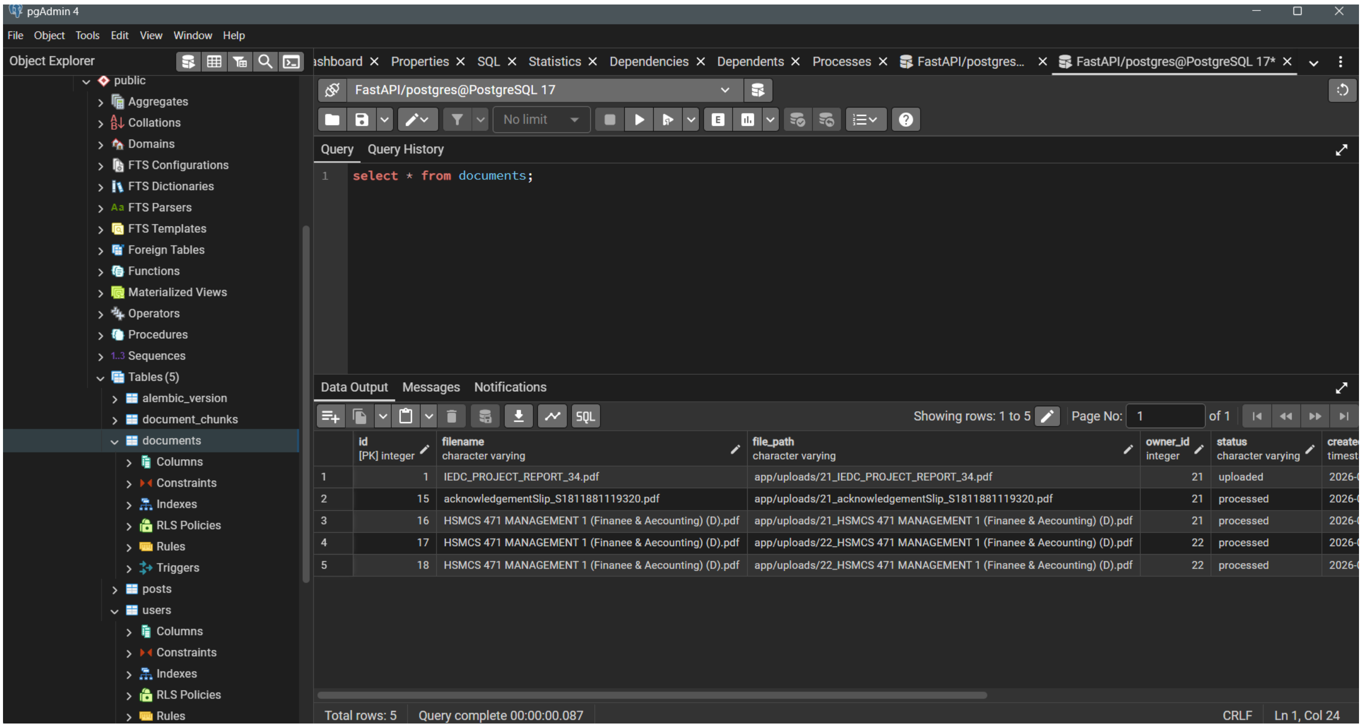

Image 1.8(Metadata of document is getting saved in Database)

6.0 : CONCLUSION & FUTURE SCOPE

6.1 Conclusion

The Document Analyzer Hub is really good at showing how we can use technologies to make it easier to look at documents. This project made a system that can look at documents that are not organized and find important information using special ways of understanding language.

The system is better than ways of searching that just look for keywords. It uses a kind of search that understands what the documents and what people are asking mean. This makes the answers more accurate and relevant. The system uses FastAPI to make sure the backend works well and PostgreSQL to store and manage information.

The system worked well throughout the project. It could look at documents quickly make codes for them and find similar documents. The results showed that the system gives answers and saves a lot of time that people would have spent looking at documents manually. It also lets people ask questions in a natural way.

The Document Analyzer Hub shows how important it is to use intelligence to deal with large amounts of unorganized information. It provides a way to look at documents that's fast and efficient which is useful in many areas, like schools, businesses and research. The Document Analyzer Hub makes it easier to work with documents. That is a big deal.

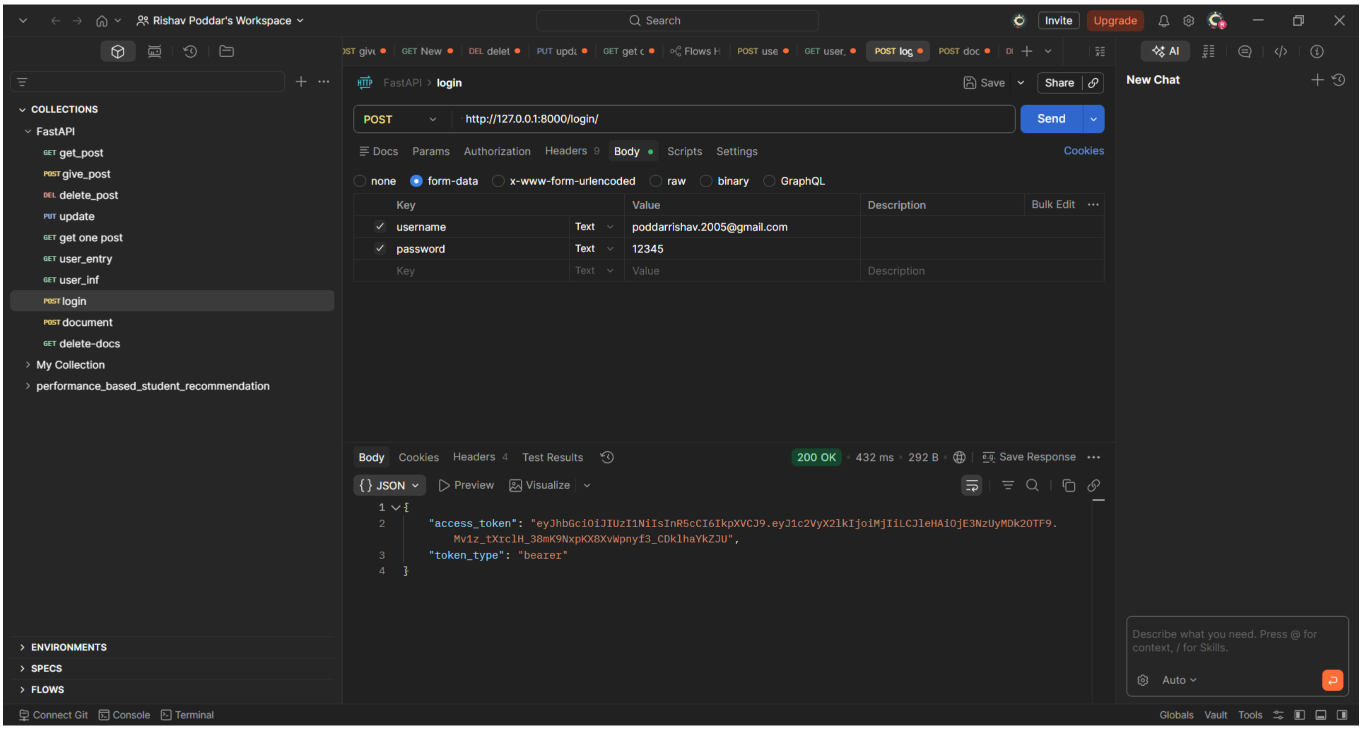

Image 1.9 (User is login and granted JWT Token)

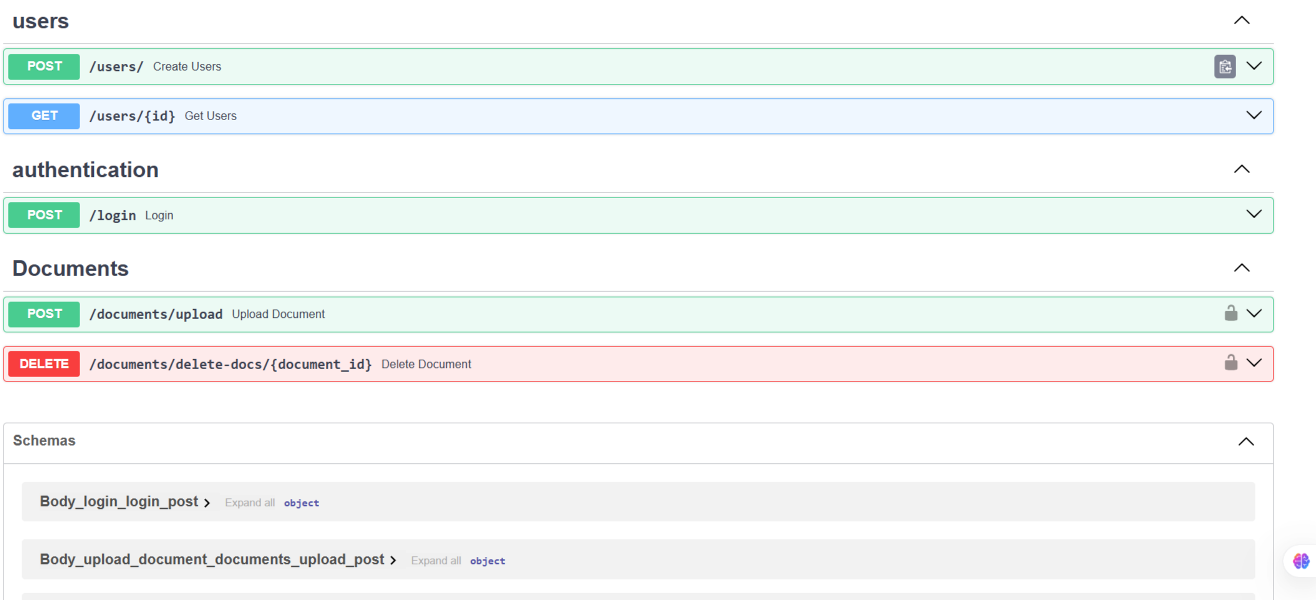

Image 1.10(API Documentation using FastAPI Swagger)

6.2 Future Scope

The current system works well for text-based documents. There are ways to make it even better. One big improvement would be to add image processing so it can extract information from scanned documents and images using Optical Character Recognition (OCR).

Another thing that could be improved is handling documents in languages. Now it mainly focuses on one language but if it could handle many languages it would be more useful worldwide.

The system could also get better by using advanced models from Hugging Face Transformers. This would make the system more accurate. Also using a mix of keyword and semantic search could help it find what you are looking for easily.

To make it handle data it could use special databases designed for large amounts of data. Putting it on a platform would also make it work better and be easier to use.

A cool new feature could be AI, which lets users chat with documents. This would make the system more friendly and useful for real-time applications.

In conclusion the AI Powered Document Analyzer Hub is a start for intelligent document processing. With improvements and new technologies it could become a powerful tool, for extracting knowledge and making decisions with AI Powered Document Analyzer Hub.

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